

Question 1 (10 marks)

- (a) Given $z_1 = 1$, $z_2 = 1+i$ and $z_3 = i$, show that $|z_2| = |z_3 - z_1|$ and that

$$\arg\left(\frac{z_3 - z_1}{z_2}\right) = \frac{\pi}{2}.$$

- (b) Sketch the region, where $\arg(z-1) \leq \frac{\pi}{3}$ and $|z-1| \geq 2$, hold simultaneously.

- (c) Given that $\sin(\theta + \alpha) = 2\sin(\theta - \alpha)$, prove $\tan \theta = 3 \tan \alpha$.

Question 2 (10 marks)

- (a) If $\frac{iz-1}{z-i}$ is real, find the value of $|z|$, where $z = a+ib$.

- (b) Find the value of $\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^{12}$, give your answer in the form $a+ib$.

- (c) Expand $\left(z + \frac{1}{z}\right)^3$, for $z = \cos \theta + i \sin \theta$, and hence show that

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$$

Question 3 (10 marks)

- (a) (i) Prove that $\sin \theta = \frac{2t}{1+t^2}$ and $\cos \theta = \frac{1-t^2}{1+t^2}$, given that $t = \tan \frac{\theta}{2}$.

- (ii) Hence prove that $\sin 2\theta = \frac{4t(1-t^2)}{(1+t^2)^2}$ and thus find the roots of the

$$\text{equation } t^4 + 8t^3 + 2t^2 - 8t + 1 = 0.$$

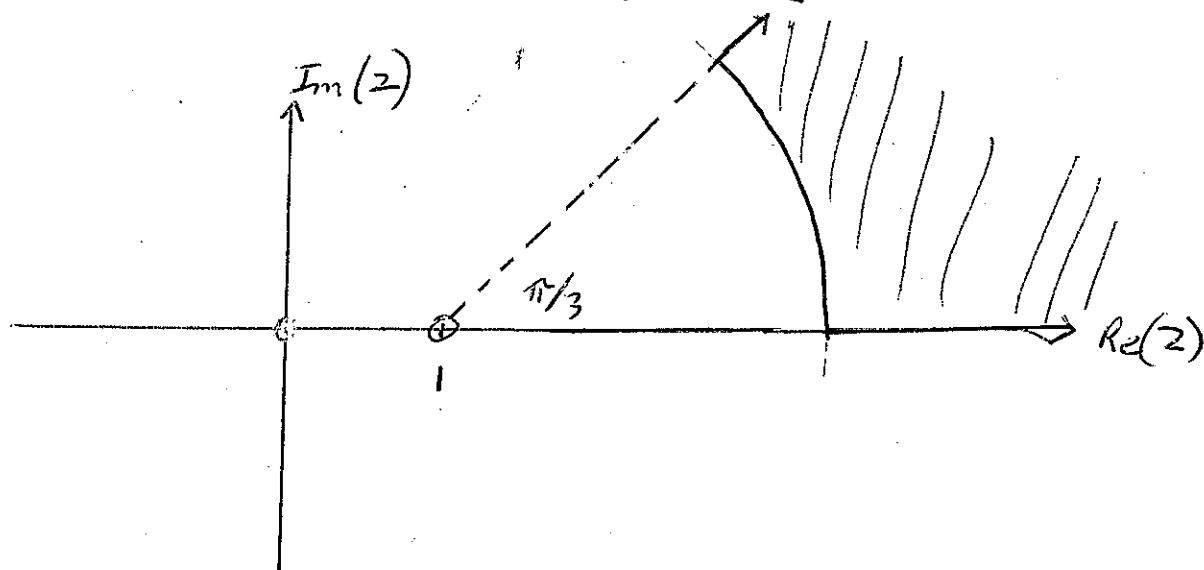
Question 1

$$a) |z_2| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$|z_3 - z_1| = |i - 1| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$$

$$\begin{aligned} \arg\left(\frac{z_3 - z_1}{z_2}\right) &= \arg\left(\frac{-1+i}{1+i}\right) = \arg(-1+i) - \arg(1+i) \\ &= \frac{3\pi}{4} - \frac{\pi}{4} = \frac{\pi}{2} \end{aligned}$$

b)



Question 2

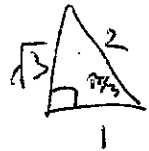
$$\begin{aligned} a) \frac{i(a+ib)-1}{a+ib-i} &= \frac{-b-1+ia}{a+i(b-1)} \times \frac{a-i(b-1)}{a-i(b-1)} \\ &= \frac{-ba+ib^2-ib-a+ib-i+ia^2+ab-a}{a^2+(b-1)^2} \\ &= \frac{-ba-a+ab-a+i(b^2-b+b+a^2-1)}{a^2+(b-1)^2} \\ &= \frac{-2a+i(a^2+b^2-1)}{a^2+(b-1)^2} \end{aligned}$$

if $\frac{i2-1}{2-i}$ is real $a^2+b^2-1=0 \therefore a^2+b^2=1$

$$|z| = \sqrt{a^2+b^2} = \sqrt{1} = 1$$

Question 2

$$b) \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \operatorname{cis} \left(\tan^{-1} \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} \right)$$



$$= \sqrt{1} \operatorname{cis} \frac{\pi}{3}$$

$$= \operatorname{cis} \frac{\pi}{3}$$

$$\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right)^{12} = \left(\operatorname{cis} \frac{\pi}{3} \right)^{12}$$

$$= \operatorname{cis} \frac{12\pi}{3} = \operatorname{cis} 4\pi = \operatorname{cis} 0$$

$$= \cos 0 + i \sin 0$$

$$= 1$$

$$c) \left(2 + \frac{1}{2} \right)^3 = \left(\operatorname{cis} \theta + \frac{1}{\cos \theta + i \sin \theta} \times \frac{\cos \theta - i \sin \theta}{\cos \theta - i \sin \theta} \right)^3$$

$$= \left(\operatorname{cis} \theta + \frac{\cos \theta - i \sin \theta}{1} \right)^3$$

$$= \left(\cos \theta + i \sin \theta + \cos \theta - i \sin \theta \right)^3$$

$$= (2 \cos \theta)^3$$

$$= 8 \cos^3 \theta$$

$$\begin{array}{ccccccc} & & 1 & & & & \\ & 1 & & 1 & & & \\ 1 & & 2 & & 1 & & \\ 1 & 3 & & 3 & 1 & & \end{array}$$

$$\left(2 + \frac{1}{2} \right)^3 = 2^3 + 3 \cdot 2^2 \left(\frac{1}{2} \right) + 3 \cdot (2) \left(\frac{1}{2} \right)^2 + \left(\frac{1}{2} \right)^3$$

$$= 2^3 + 3 \cdot 2 + 3 \cdot 2^{-1} + 2^{-3}$$

$$= (\cos \theta + i \sin \theta)^3 + 3(\cos \theta + i \sin \theta) + 3(\cos \theta + i \sin \theta)^{-1} + (\cos \theta + i \sin \theta)^{-3}$$

$$= \cos 3\theta + i \sin 3\theta + 3\cos \theta + i 3\sin \theta + 3\cos(-\theta) + i 3\sin(-\theta) + \cos(-3\theta) + i \sin(-3\theta)$$

$$= \cos 3\theta + i \sin 3\theta + 3\cos \theta + i 3\sin \theta + 3\cos \theta - i 3\sin \theta + \cos 3\theta - i \sin 3\theta$$

$$= 2 \cos 3\theta + 6 \cos \theta$$

$$\therefore 8 \cos^3 \theta = 2 \cos 3\theta + 6 \cos \theta$$

$$2 \cos 3\theta = 8 \cos^3 \theta - 6 \cos \theta$$

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$$